Kubernetes Practical Assignment

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Container Orchestration
Tool Used: Kubernetes (Minikube)

Kubernetes Assignment - 1

Tasks Performed

1. Deploy a Kubernetes cluster for 3 nodes
2. Create a NGINX deployment with 3 replicas

1.1 Kubernetes Cluster Setup

A Kubernetes cluster was created using **Minikube**.

Start cluster:

minikube start

Verify cluster nodes:

kubectl get nodes

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   3h24m   v1.35.1
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$
```

kubectl get nodes

1.2 Create NGINX Deployment

Deployment YAML:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 replicas: 3
 selector:
   matchLabels:
     app: nginx
 template:
   metadata:
     labels:
       app: nginx
   spec:
     containers:
     - name: nginx
       image: nginx
       ports:
       - containerPort: 80
```

Apply deployment:

kubectl apply -f deployment.yaml

Verify replicas:

kubectl get deployments

Check pods:

kubectl get pods

📸 Screenshots

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   3/3     3            3           14s
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$
```

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75fdcbbc74-h78hp   1/1     Running   0          38s
nginx-deployment-75fdcbbc74-ls7xz   1/1     Running   0          39s
nginx-deployment-75fdcbbc74-rmh4g   1/1     Running   0          39s
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$
```

kubectl get deployments

kubectl get pods

Kubernetes Assignment - 2

Tasks Performed

1. Use previous deployment
2. Create NodePort service for NGINX deployment
3. Verify service on browser

Create NodePort Service

Service YAML:

```
apiVersion: v1
kind: Service
metadata:
 name: nginx-service
spec:
 type: NodePort
 selector:
   app: nginx
 ports:
 - port: 80
   targetPort: 80
   nodePort: 30007
```

Apply service:

```
kubectl apply -f service-nodeport.yaml
```

Verify service:

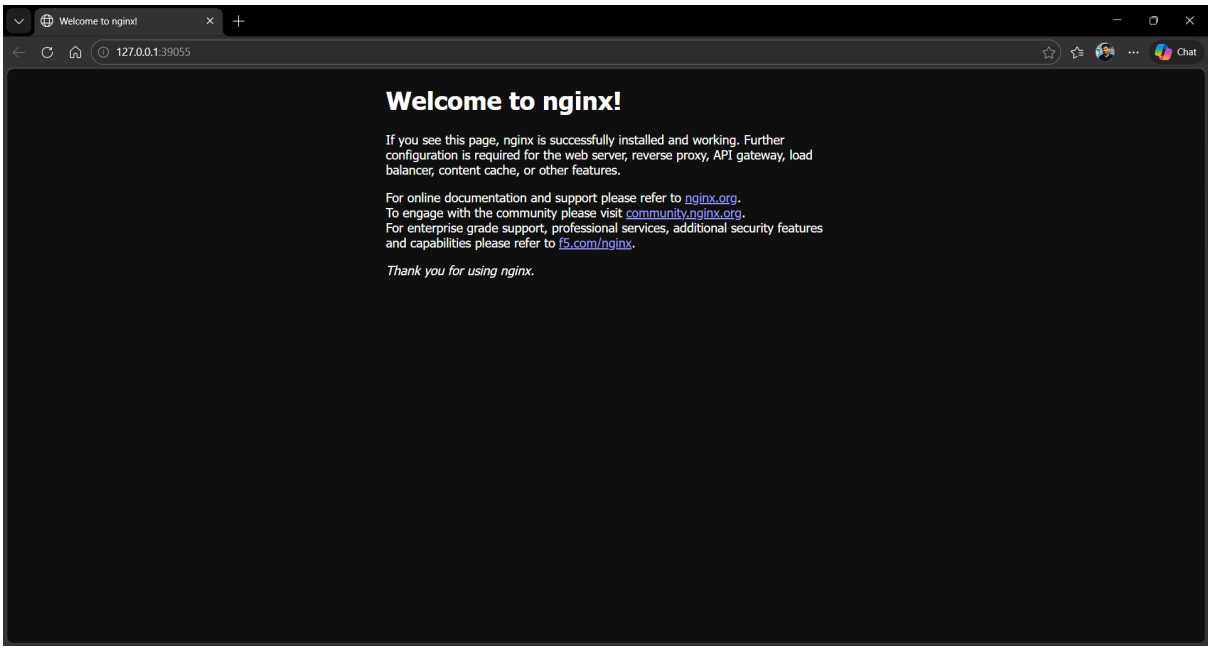
```
kubectl get services
```

Open service:

```
minikube service nginx-service
```

📸 **Screenshots**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ minikube service nginx-service
|-----------|-------------|-------------|---------------------------|
| NAMESPACE |    NAME     | TARGET PORT |           URL             |
|-----------|-------------|-------------|---------------------------|
| default   | nginx-service | 80          | http://192.168.49.2:30007 |
|-----------|-------------|-------------|---------------------------|
🔗  Starting tunnel for service nginx-service.
|-----------|-------------|-------------|---------------------------|
| NAMESPACE |    NAME     | TARGET PORT |           URL             |
|-----------|-------------|-------------|---------------------------|
| default   | nginx-service |             | http://127.0.0.1:39055    |
|-----------|-------------|-------------|---------------------------|
🎉  Opening service default/nginx-service in default browser...
/usr/bin/xdg-open: 882: x-www-browser: not found
/usr/bin/xdg-open: 882: firefox: not found
/usr/bin/xdg-open: 882: iceweasel: not found
```



kubectl get services
minikube service nginx-service

Kubernetes Assignment - 3

Task

Change replicas to **5**

Edit deployment:

kubectl scale deployment nginx-deployment --replicas=5

Verify:

kubectl get pods

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75fdcbbc74-9p9w2   1/1     Running   0          13s
nginx-deployment-75fdcbbc74-ccwtj   1/1     Running   0          13s
nginx-deployment-75fdcbbc74-h78hp   1/1     Running   0          3m52s
nginx-deployment-75fdcbbc74-ls7xz   1/1     Running   0          3m53s
nginx-deployment-75fdcbbc74-rmh4g   1/1     Running   0          3m53s
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$
```

kubectl get pods

Kubernetes Assignment - 4

Task

Change service type to **ClusterIP**

Edit service YAML:

```
apiVersion: v1
kind: Service
metadata:
 name: nginx-service
spec:
 type: ClusterIP
 selector:
   app: nginx
 ports:
 - port: 80
   targetPort: 80
```

Apply:

kubectl apply -f service-clusterip.yaml

Verify:

kubectl get services

📸 **Screenshot**

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get services
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes      ClusterIP   10.96.0.1       <none>        443/TCP   3h32m
nginx-service   ClusterIP   10.109.39.134   <none>        80/TCP    4m8s
```

kubectl get services

Kubernetes Assignment - 5

Tasks

1. Use previous deployment
2. Deploy another NGINX deployment with 3 replicas
3. Create ClusterIP service
4. Configure ingress service

Create Second Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-app2
spec:
 replicas: 3
 selector:
   matchLabels:
     app: nginx-app2
 template:
   metadata:
     labels:
       app: nginx-app2
   spec:
     containers:
     - name: nginx
       image: nginx
       ports:
```

```
      - containerPort: 80
```

Apply:

```
kubectl apply -f nginx-app2.yaml
```

Create Service

```
apiVersion: v1
kind: Service
metadata:
 name: nginx-app2-service
spec:
 type: ClusterIP
 selector:
   app: nginx-app2
 ports:
 - port: 80
   targetPort: 80
```

Apply:

```
kubectl apply -f nginx-app2-service.yaml
```

Create Ingress

First enable ingress:

```
minikube addons enable ingress
```

Ingress YAML:

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: nginx-ingress
spec:
 rules:
 - http:
     paths:
     - path: /nginx1
       pathType: Prefix
       backend:
         service:
           name: nginx-service
           port:
```

```
          number: 80
    - path: /nginx2
      pathType: Prefix
      backend:
        service:
          name: nginx-app2-service
          port:
            number: 80
```

Apply:

kubectl apply -f ingress.yaml

Verify:

kubectl get ingress

📸 Screenshots

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-app2         3/3     3            3           3m29s
nginx-deployment   5/5     5            5           11m
```

kubectl get deployments

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get services
NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes           ClusterIP   10.96.0.1       <none>        443/TCP   3h38m
nginx-app2-service   ClusterIP   10.106.226.34   <none>        80/TCP    2m46s
nginx-service        ClusterIP   10.109.39.134   <none>        80/TCP    10m
```

kubectl get services

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get ingress
NAME            CLASS   HOSTS   ADDRESS         PORTS   AGE
nginx-ingress   nginx   *       192.168.49.2    80      49s
```

kubectl get ingress

```
vishal@LAPTOP-URMIDNIG:/mnt/d/PW-Assignments/kubernetes2-assignment$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-app2-5c597bdb78-lzgdx         1/1     Running   0          3m49s
nginx-app2-5c597bdb78-t6nrs         1/1     Running   0          3m49s
nginx-app2-5c597bdb78-x2zsp         1/1     Running   0          3m49s
nginx-deployment-75fdcbbc74-9p9w2   1/1     Running   0          8m39s
nginx-deployment-75fdcbbc74-ccwtj   1/1     Running   0          8m39s
nginx-deployment-75fdcbbc74-h78hp   1/1     Running   0          12m
nginx-deployment-75fdcbbc74-ls7xz   1/1     Running   0          12m
nginx-deployment-75fdcbbc74-rmh4g   1/1     Running   0          12m
```

kubectl get pods